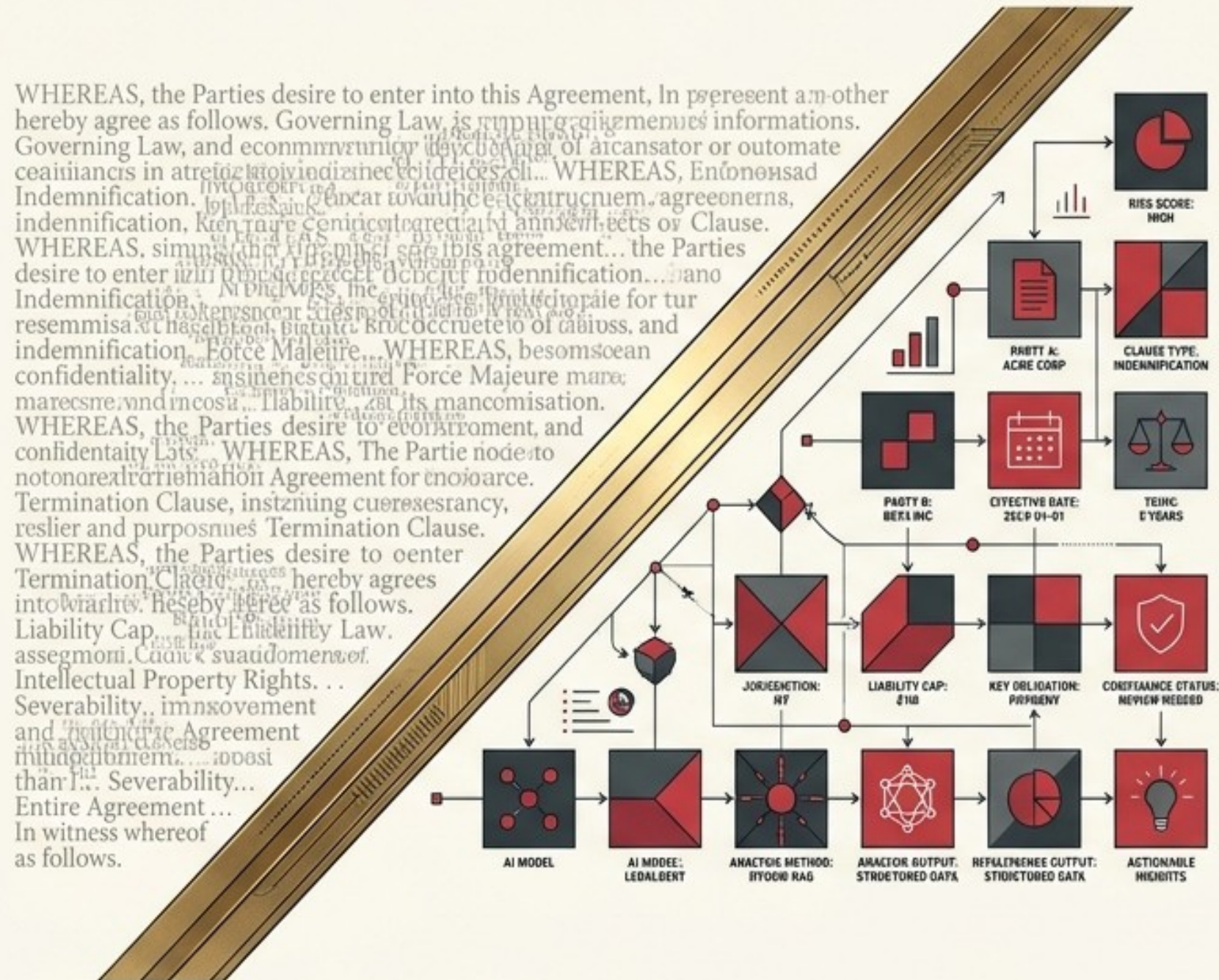


AI Legal Document Intelligence Platform

Transforming Contract Analysis with Hybrid RAG, LegalBERT, and LLMs



Simplifying Legal Understanding

Manual Review



Lengthy, complex contracts
prone to human oversight.

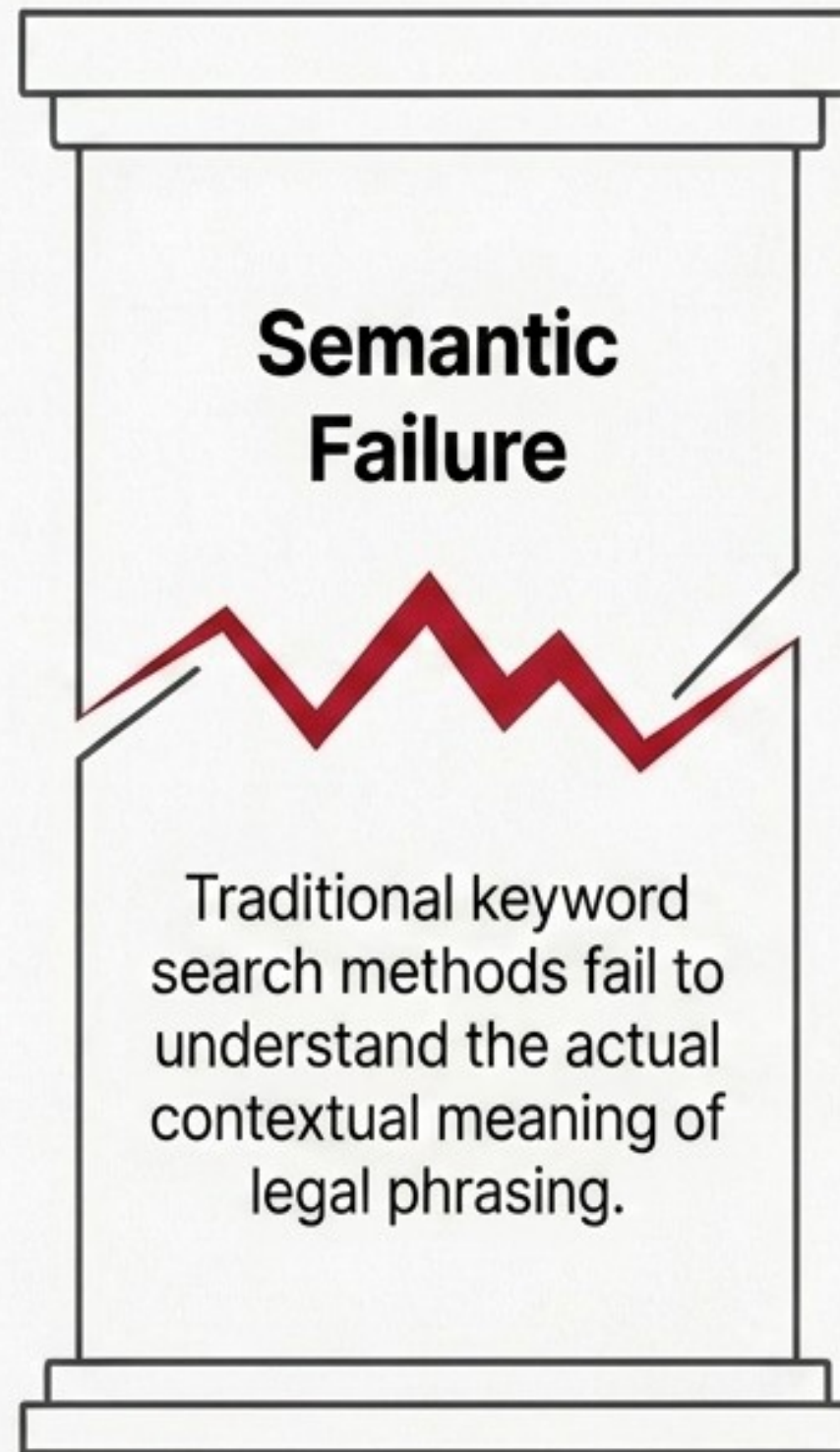
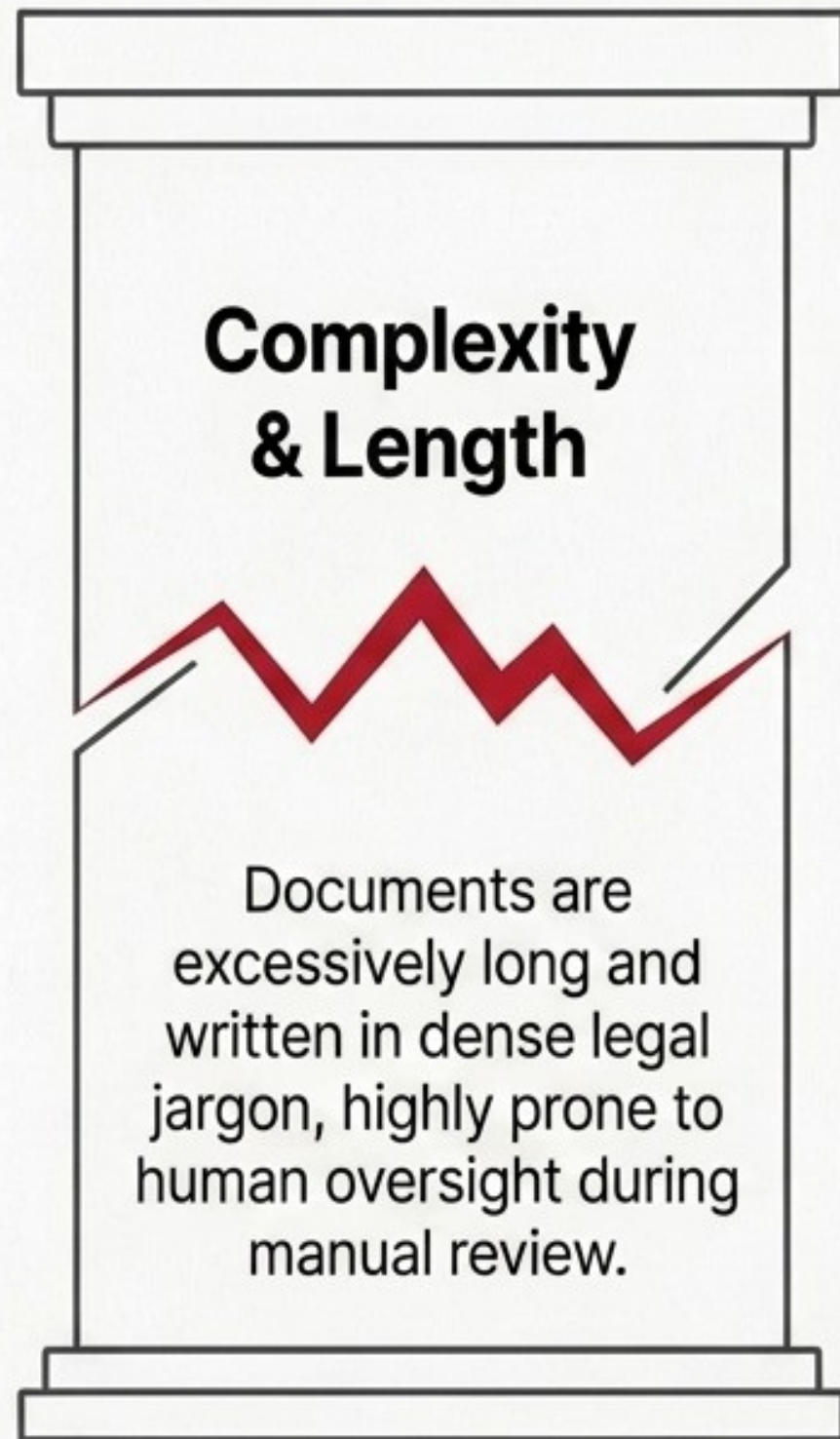
AI-Powered Insights



Instant, automated extraction
replacing time-consuming reading.

Core Capabilities: Clause classification, risk assessment, obligation extraction, and conversational Q&A.

The Challenge of Legal Text



System Objectives



Intelligent Automation

Drastically reduce review time without sacrificing the precision required for legal documentation.



Advanced Classification

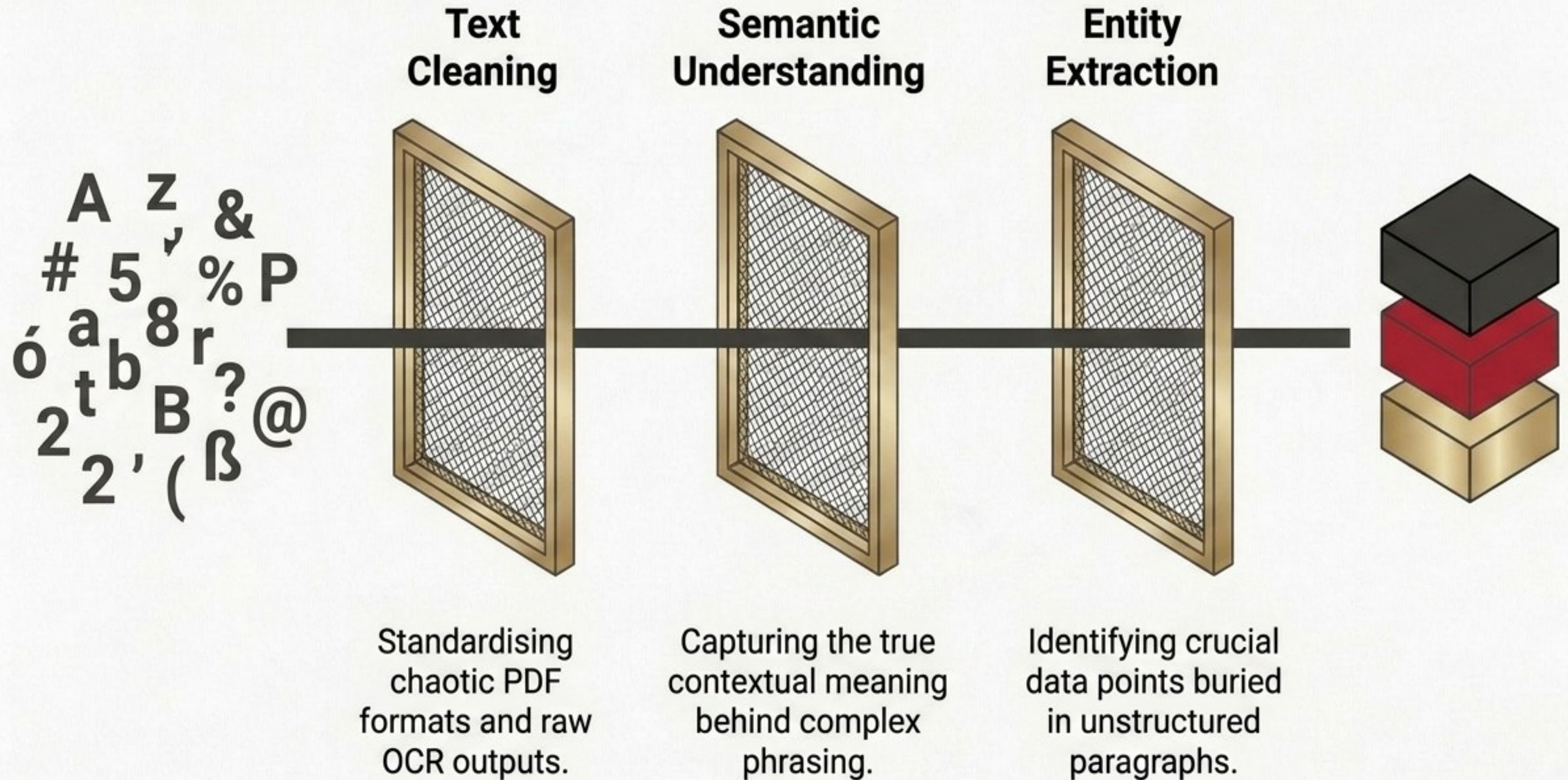
Programmatically categorise clauses, identify risks, and ensure compliance across hundreds of pages.



Grounded Q&A

Provide accurate, explainable answers rooted strictly in the provided document text to eliminate hallucination.

The Role of Natural Language Processing



A Modern, Modular Tech Stack

Backend & Storage

- Python, FastAPI, SQLite, Vector Database (Parquet Embeddings).

AI & NLP

- Transformers, Hugging Face, PyTorch, RAG architecture, OCR integration.

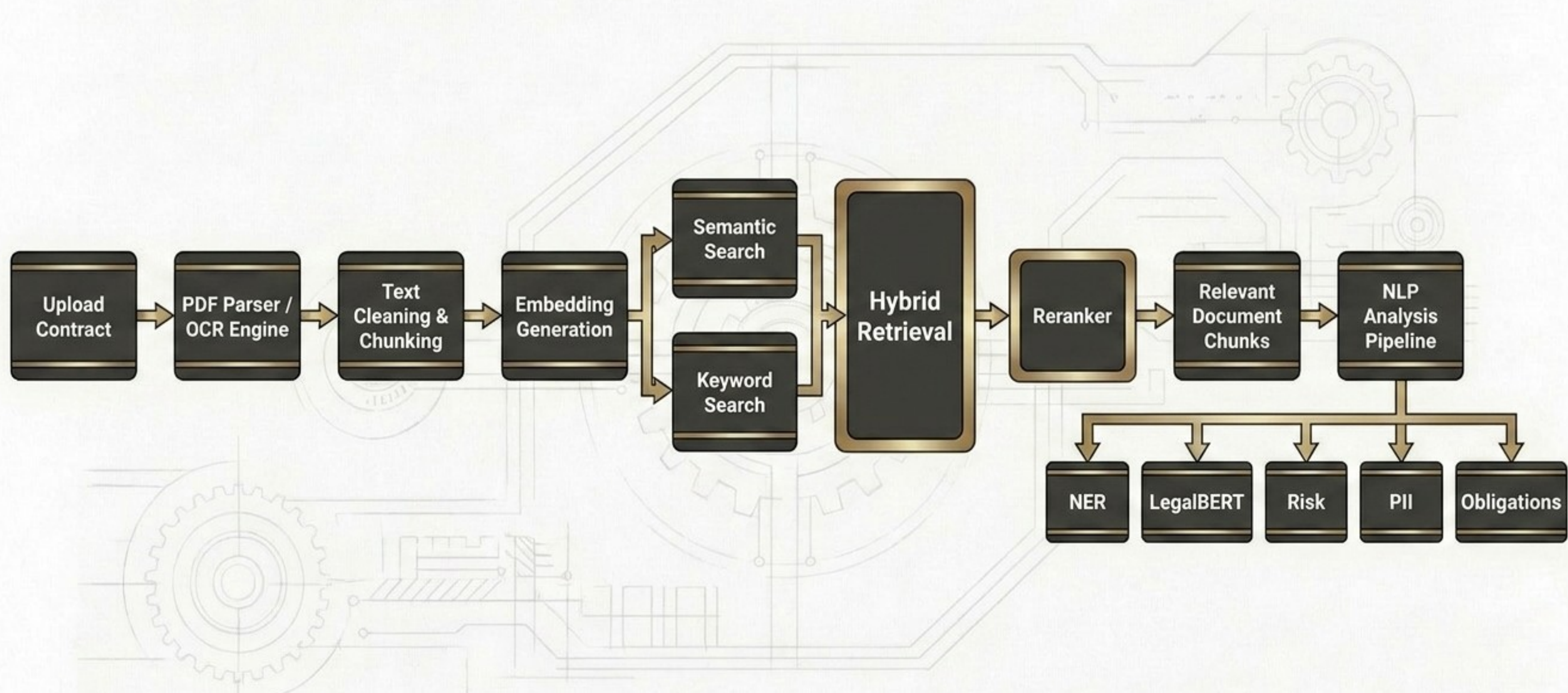
Domain-Specific Models

- LegalBERT, custom Reranking pipelines.

Large Language Models

- Groq and Ollama (Flexible inference providers).

The Data Pipeline



Deep Dive: LegalBERT

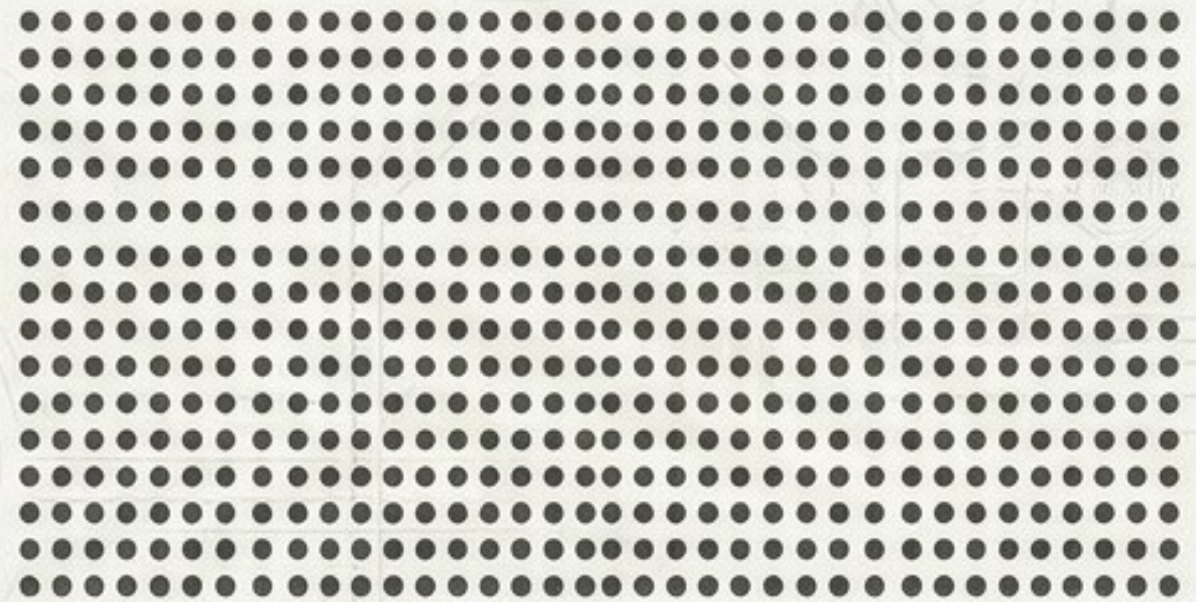
General Purpose Model



Struggles with nuanced legal terminology and contextual meaning.

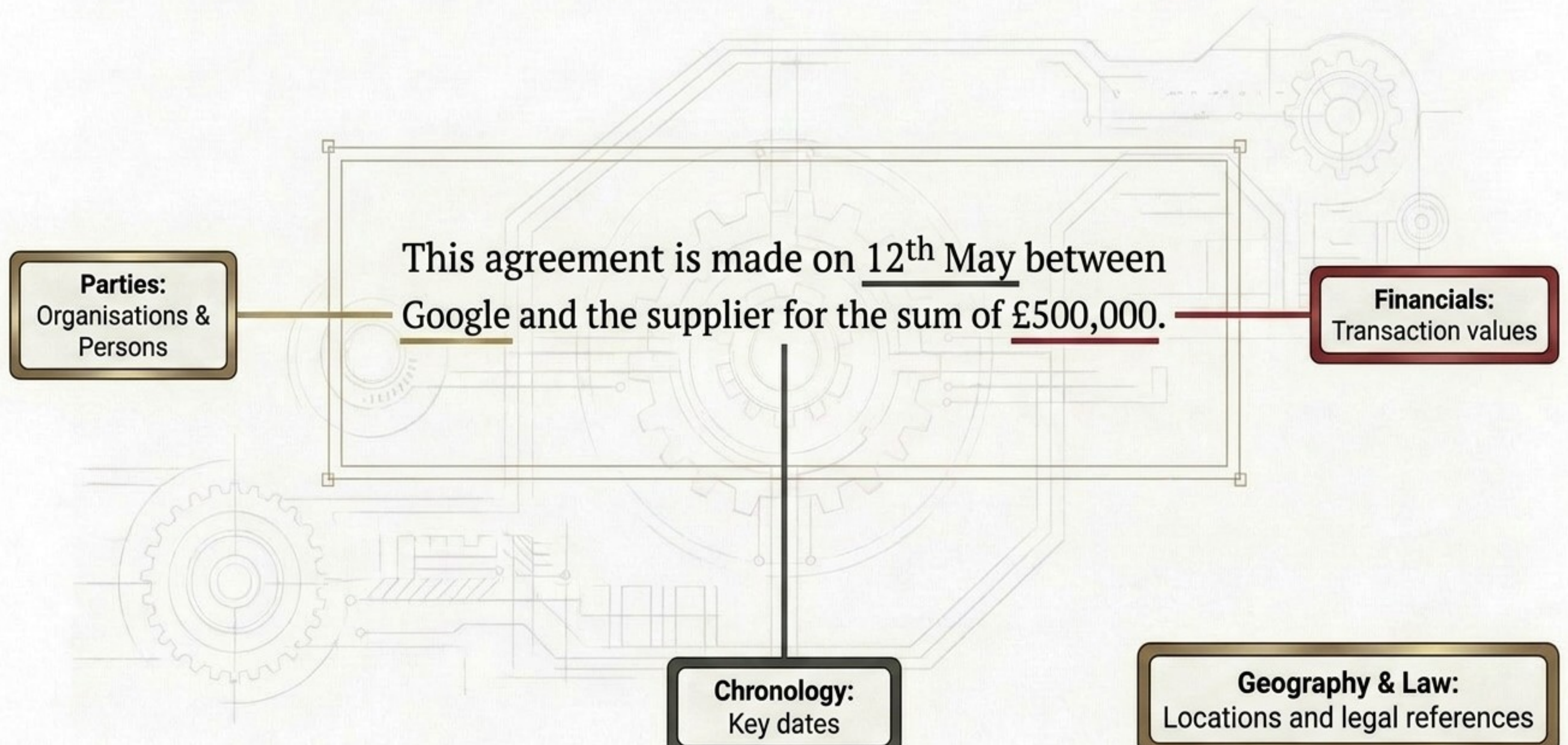
LegalBERT

Domain
Mastery

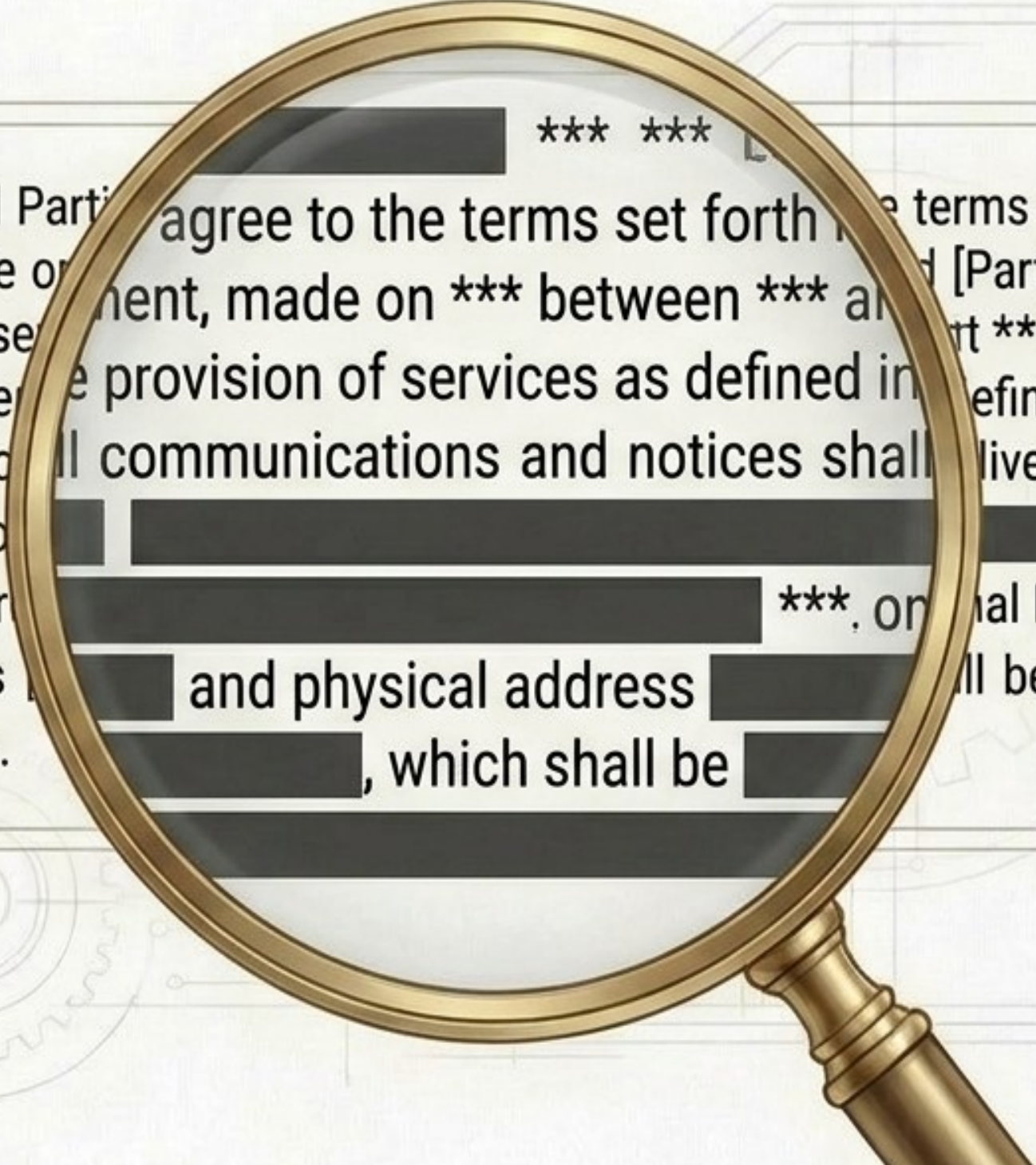


- **Pretrained on Law:** Specifically trained exclusively on legal documents and terminology.
- **Clause Classification:** Outperforms general models in accurately categorising legal language.
- **Context Representation:** Captures precise semantic meaning (e.g., 'consideration' as financial value).

Extracting Crucial Context (NER)



Privacy and Data Protection



The undersigned Party agree to the terms set forth in the terms set forth in this agreement, made on *** between *** and [Party B], relating to the provision of services as defined in Section 2. All communications and notices shall be delivered via email to [email address] of [redacted]. Confidential information, including but not limited to, ***, original ID [redacted] and physical address [redacted] and physical address [redacted] will be maintained in strict confidence. [redacted], which shall be [redacted].

Identify & Protect:

Automated detection and redaction of sensitive data streams.

Targeted Entities:

Names, email addresses, and phone numbers

Regulatory Compliance:

Identification of National IDs and physical addresses

Automated Risk Assessment

Low Risk

Standard boilerplate text requiring minimal oversight.

Medium Risk

Ambiguous terms or unusual phrasing requiring manual review.

High Risk

Uncapped liabilities, severe penalties, or critical compliance breaches demanding immediate attention.

Predictive Analysis Workflow

Rule-Based Compliance Engine

Liability Caps Present



Data Protection Clause



Missing: Force Majeure Clause



- **Automated Auditing:** Checks uploaded contracts against predefined legal rules.
- **Gap Analysis:** Identifies critically missing clauses before signing.
- **Risk Mitigation:** Proactively flags potentially non-compliant language.

Clarifying Responsibilities

Raw Contractual Text

The supplier shall deliver the products within 30 days.

Obligation
Extraction

Obligation: Deliver products within 30 days.

- **Actionable Insights**
isolates specific duties
- **Clarity** strips away jargon
- **Tracking** enables seamless export.

Grounded Reasoning and Q&A

- **Flexible Providers:**

Groq and Ollama inference.

- **Core Tasks:**

Summarisation and AI assistance.

- **The NLP Advantage:**

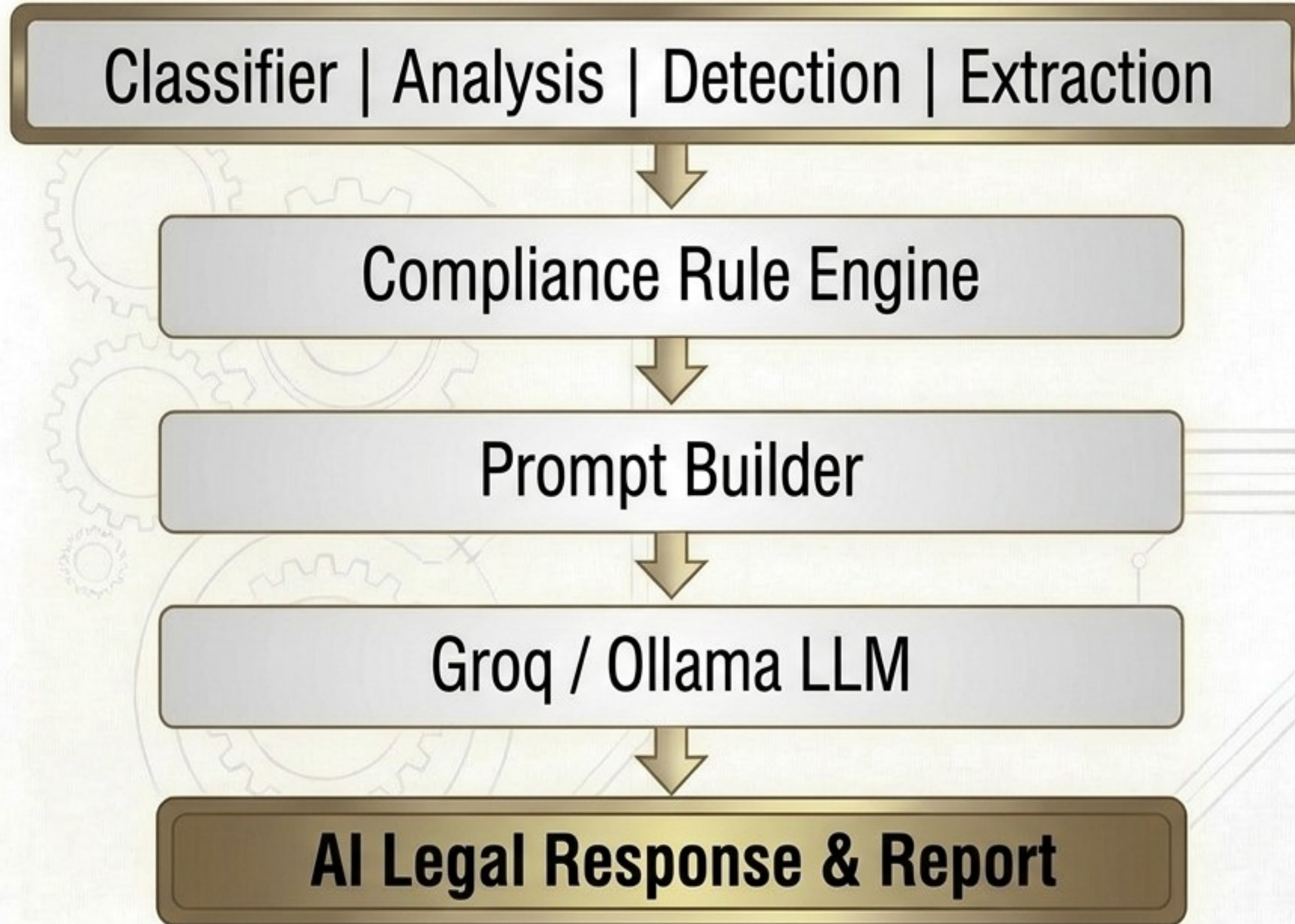
LLMs utilise structured RAG outputs for hallucination-free responses.

User Prompt: What are the termination conditions?

AI Response: Termination occurs upon 30 days' written notice by either party for material breach, insolvency, or upon completion of the project as defined in the statement of work, unless extended by mutual agreement.

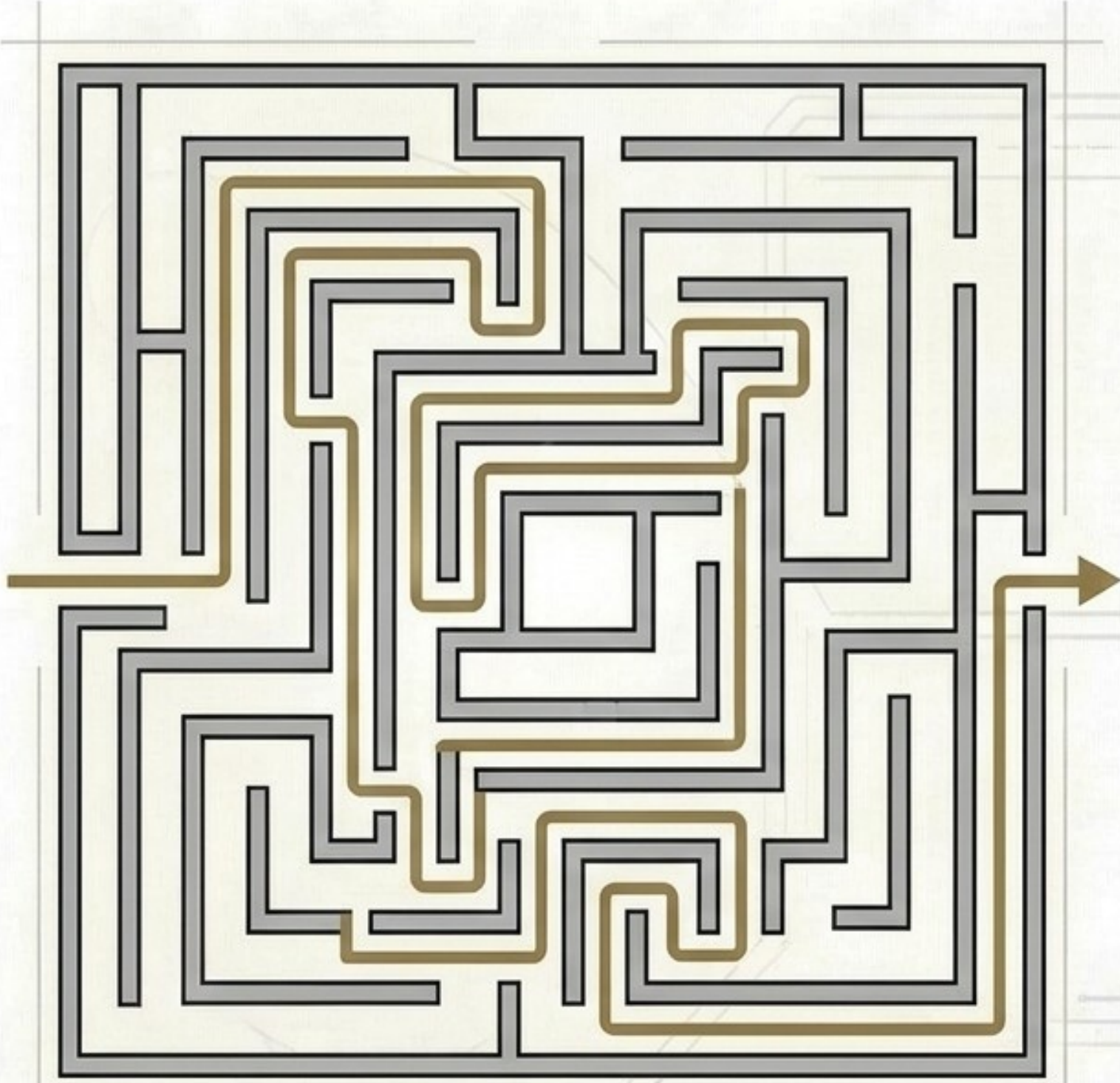
Source: Clause 4.2

The Complete Analysis Workflow



- **Sequential Synergy** feeds structured data forward.
- The **Prompt Builder** weaves context together for a definitive, accurate report.

Navigating Development Hurdles



Accuracy vs. Nuance: Balancing precise automated extraction with the subjective nature of legal jargon.

Hallucination Mitigation: Developing Hybrid RAG architecture to ensure the LLM strictly adheres to context.

OCR Limitations: Engineering robust pipelines to handle poor-quality scans and formatting inconsistencies.

Expanding the Platform's Horizons



The Future of Legal Intelligence



Transformative Efficiency: Replacing hours of manual review with instant analysis.

Technological Synergy: Merging Hybrid RAG, LegalBERT, and modern LLMs.

Real-World Impact: Delivering insights that are faster, safer, and grounded in truth.